

| Question             |                       |                       | Marking details                                                                                                                                                                                                                                                            |                       |                     |                       |                      | Marks available       |            |                       |        |        |        |  |  |   |   |   |  |  |
|----------------------|-----------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------|-----------------------|----------------------|-----------------------|------------|-----------------------|--------|--------|--------|--|--|---|---|---|--|--|
|                      |                       |                       |                                                                                                                                                                                                                                                                            |                       |                     |                       |                      | AO1                   | AO2        | AO3                   | Total  | Maths  | Prac   |  |  |   |   |   |  |  |
| 7                    | (a)                   | (i)<br>(ii)<br>(iii)  | The same (1)<br>Increases (1)<br>Increases (1)                                                                                                                                                                                                                             |                       |                     |                       |                      | 1<br>1                |            | 1                     | 3      |        |        |  |  |   |   |   |  |  |
|                      | (b)                   | (i)                   | <table><tr><td>Initial speed (km/h)</td><td>Initial speed (m/s)</td><td>Thinking distance (m)</td><td>Braking distance (m)</td><td>Stopping distance (m)</td></tr><tr><td>60</td><td>16.8 (1)<br/>Accept 17</td><td>25 (1)</td><td>20 (1)</td><td>45 (1)</td></tr></table> | Initial speed (km/h)  | Initial speed (m/s) | Thinking distance (m) | Braking distance (m) | Stopping distance (m) | 60         | 16.8 (1)<br>Accept 17 | 25 (1) | 20 (1) | 45 (1) |  |  | 4 | 4 | 4 |  |  |
| Initial speed (km/h) | Initial speed (m/s)   | Thinking distance (m) | Braking distance (m)                                                                                                                                                                                                                                                       | Stopping distance (m) |                     |                       |                      |                       |            |                       |        |        |        |  |  |   |   |   |  |  |
| 60                   | 16.8 (1)<br>Accept 17 | 25 (1)                | 20 (1)                                                                                                                                                                                                                                                                     | 45 (1)                |                     |                       |                      |                       |            |                       |        |        |        |  |  |   |   |   |  |  |
|                      |                       | (ii)                  | Substitution into: $s = \frac{d}{t}$ <b>ecf</b> (1)<br><br>Manipulation i.e. $t = \frac{25}{16.8}$ (1)<br><br>Time = 1.488 [s] or 1.49 [s] or 1.5 [s] (1)<br>N.B. Use of 17 m/s → 1.47 [s]                                                                                 |                       |                     |                       |                      | 1                     | 1<br><br>1 |                       | 3      | 2      |        |  |  |   |   |   |  |  |
|                      |                       | (iii)                 | $v = 0$ (1)<br>Substitution: $= 16.8^2$ <b>ecf</b> + $(2 \times a \times 20)$ <b>ecf</b> (1)<br>$a = [-]7.056$ [m/s <sup>2</sup> ] (1)<br>N.B. Use of $u = 17$ m/s → $[-] 7.225$ [m/s <sup>2</sup> ] accept correctly rounded answers                                      |                       |                     |                       |                      | 1                     | 1<br>1     |                       | 3      | 3      |        |  |  |   |   |   |  |  |
|                      | (c)                   | (i)                   | Seat belts produce an <u>unbalanced force</u> on the occupants (1)<br>which prevents them continuing moving forward [at a constant speed] / so the occupant remains in their seat (1)<br>Award 1 mark for N1L correctly stated if no other creditworthy response provided  |                       |                     |                       |                      | 2                     |            |                       | 2      |        |        |  |  |   |   |   |  |  |